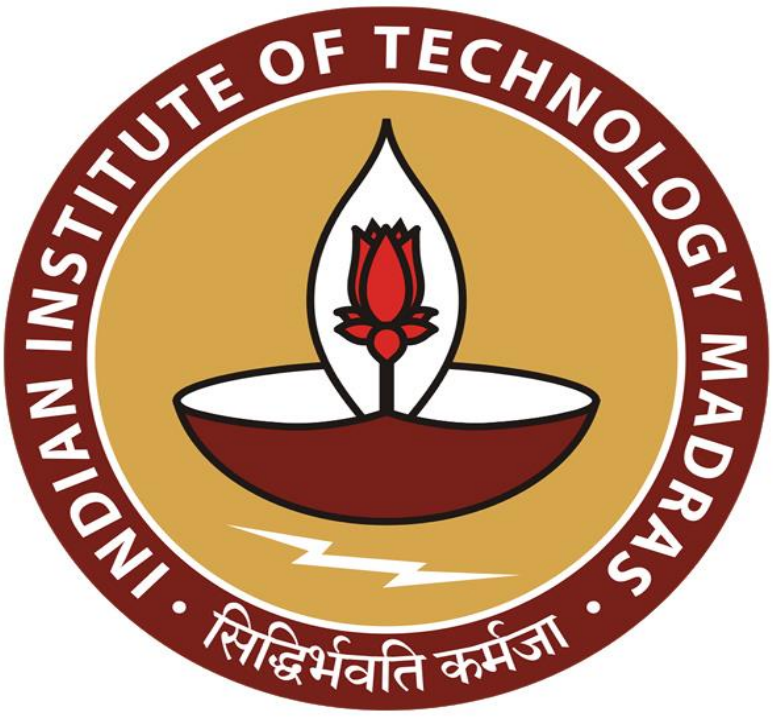
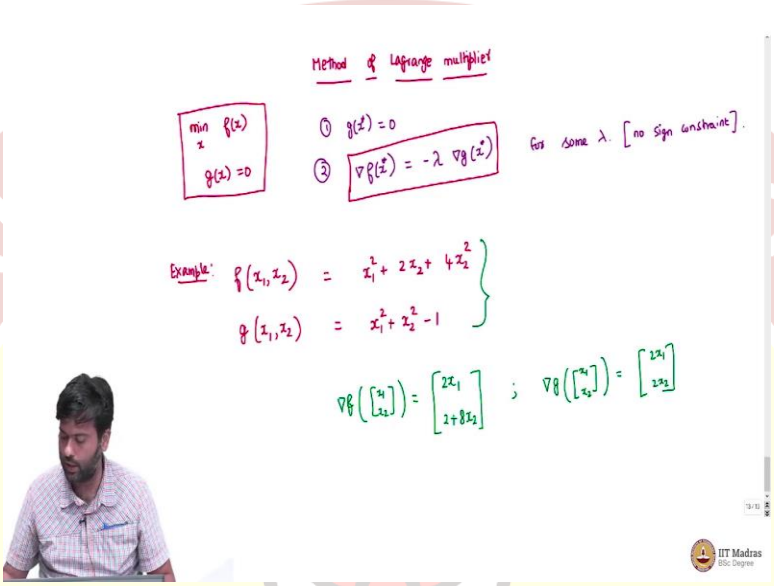


IIT Madras

ONLINE DEGREE

Projected Gradient Descent
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Welcome back and what we are going to do now is looked at the method of Lagrange multipliers specifically to solve problem, which are constrained optimization problems with the equality constraint. So, we are of the form minimize over x f(x) such that g(x) equals 0. We saw earlier that to solve this problem and necessary condition that optimal solution should satisfy is that, of course, g(x*) should be 0 and secondly, we also argued that ∇f(x*) should be - λ times ∇of g(x*) for some λ.

So, no sign constraint, it can be either positive or negative. This has to be satisfied. Now, let us see how we can use this insight that the gradients have to align either in parallel or anti parallel for x star to be some optimal point to actually solve a simple problem. So, we will take a simple example and try to solve this using the method of Lagrange multipliers.

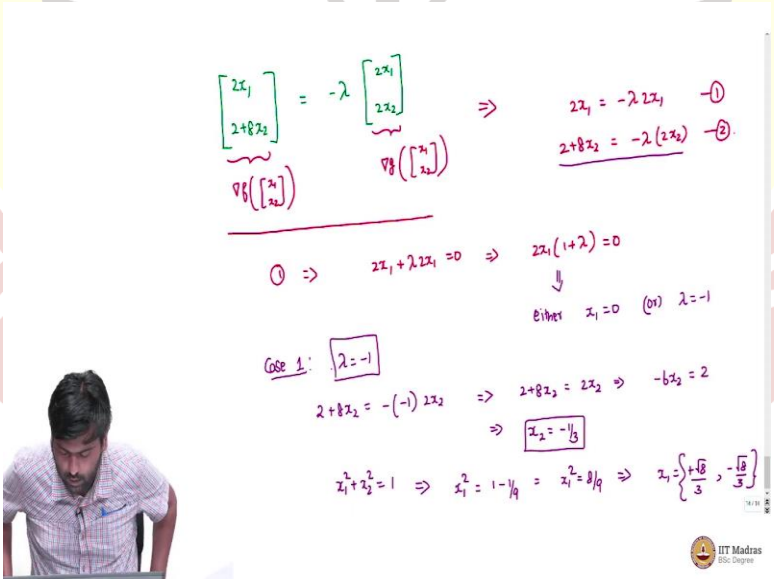
So, let us take a example where we have a function example. This is a function of 2 variables, let us say x1 and x2 and the function looks like this. So, x1 squared + 2x2 + 4, let us say x2 squared. This is just a quadratic function, a simple quadratic function and now we want to argue that we somehow want to also of course, I need to also say what is the g, let us say g of x1 , x2 is x1 squared + x2 squared, let us say - 1.

So, which means that we want to find an x1, x2 which is on the unit circle, such that because g of x1, x2 should be 0, which means that x1 squared + x2 squared equals 1, which let us say minimize as x1 squared + 2x2 + 4x2. So, this is the problem that we want to solve. Now, how do we solve this? Our Lagrange multiplier insight tells us that a necessary condition is that ∇f(x*)should be - λ ∇g(x*). Now, how can we exploit this condition? Let us, try to compute for any x what is ∇f(x)and ∇g(x).

So, in this for this particular specific problem that we have a had. So, for any x ∇f of some x1 x2 would be what in this case, it is simply the partial derivative with respect to x1 in the first component and partial derivative with respect to x2 in the second component which would happen to be let us say 2x1 and 2 + 8x2, this would be the gradient with respect to x1 x2 at any point x1 x2.

Similarly, the gradient with respect to the constraint function g at x1, x2 is going to be something like maybe in this case, it will be 2x1, 2x2. Now, we know by the method of Lagrange multipliers that the optimal should necessarily satisfy something which is equation 2, which means that gives us some set of equations here.

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So, what equations does it gives us it gives us that 2x1 which was the gradient of f and 2 + 8x2. Now, this should be - λ times 2x1 2x2 which is ∇g let me just note it down so that we are clear what we are talking about. This is ∇f of x1 x2 maybe I should at x1 x2 and this is the ∇g at x1 x2.

But now, this simply says that we have a system of equations. So, what is the equation so 2x1 should be - λ times 2x1 that is equation number 1 and 2 + 8x2 equals - λ times 2x2 that is equation number 2.

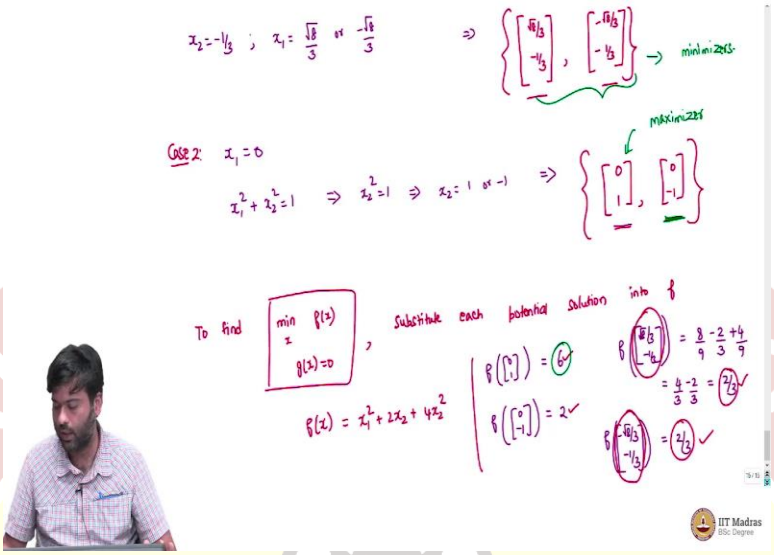
So, let us look at the equation number 1, 1 implies 2x1 + λ times 2x1 equals 0, which implies 2x1 into (1 + λ) equals 0. So, this there should be this should necessarily happen there are 2 ways this could happen. The first way is, so this implies either x1 is 0, for which this might be true or λ equals - 1.

When in which case also this is true. Let us, maybe first look at the case 1, when λ equals - 1. So, let us assume λ is - 1. If λ is - 1, then how can we exploit the fact that if λ is - 1, what can we say we can say, now we can look at equation 2 and say that 2 + 8x2 should be equal to - λ, which is - 1 times 2x2, which implies 2 + 8x2 now equals 2x2, which further implies - 6x2 equals 2. Now, that gives us a value for x2 as - 1 by 3.

So, if you know that λ is - 1, which is case 1 that we are looking at, then x2 necessarily has to be - 1 by 3. But how do we get x1, so if I know x2, then I can simply get x1, because we necessarily have to satisfy not just the Lagrange equation, but also the feasibility equation. Which is x1 squared + x2 squared should be equal to 1, this also has to be satisfied.

Now, we have found x2 now from which we can simply get x1. So, x1 squared, then would be 1 - x2 squared, which in this case is 1 over 9 which means that x1 squared is 8 over 9, which means x1 is either + square root of 8 by 3 or - square root of 8 by 3. So, there are 2 options for x1. So case 1, when λ equals - 1 gives us 2 possible potential solutions.

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Which are as follows. We found x2 as - 1 by 3 and x1 as either square root of 8 by 3, or - square root of 8 by 3. So, which implies the points that are important here are x1 x2, which means that it is square root of 8 by 3 - 1 by 3 , - square root of 8 by 3, , - 1 by 3. We have 2 sets of points, 2 set of 2 points.

So, this is remember case when we argued that λ is - 1. The other case is it could so happen that x1 was 0. So, there are 2 cases that we got from the first equation 2, 2x1 times 1 + λ is 0 was the equation and that can either be satisfied if x1 is 0, or if λ is - 1. We have already looked at the case where λ is - 1 now if x1 is 0, now how do we get x2. Now we look at the feasibility equation. So, the feasibility equation would say, x1 squared + x2 squared equals 1, which implies if x1 is 1, then x2 squared has to be 1, which means x2 equals 1 or - 1.

Now, this then tells us the set of points which are important here are the following. Which should be x1 is 0 and x2 is 1 and x1 is 0 and x2 is - 1. So now, we are saying that if it is optimal then necessarily it has to satisfy a certain equation, which means that there are only 4 possible options for the optimal point. So, now to find the optimal point so to find min over x f(x) such that g(x) is 0, in this case. What we have done is substitute each potential solution into f.

So, you have 4 possibilities and now we substitute each of these possibilities into f and see which one gives us the least value. Now, let us go ahead and do that for completions purpose. So, what

was our f (x)? Our f(x) remember was, let me go back to make sure (())(10:57) right f(x) x1 squared + 2x2 + 4 x2 square.

Now, what does this tell us? This tells us that if f of let us do the simple things first 0, 1, f of 0, 1 is what? It is simply 2 + 4, which is 6 and f of 0 - 1 is again, 2 not. So, it will be - 2 + 4, which is 2. These are the 2 values now what about f of square root of 8 by 3, - 1 by 3, that us going to be 8 by 9 - 2 by 6 2x2 x2 is - 1 by 3, + 4 times x2 squared, which is 4 over 9.

So, which is 12 over 9 - 1 over 3, which is 4 by 3 which is let me do make sure I am doing this correct. 4 over 3 - 1 by 3 this is 1, if I am not wrong. I think I am right sorry I think it is not 6 it is 3. So, that could give me 4 by 3 - 2 by 3, which will be 2 over 3. So, that would be the third value, and the other 1 that is left is - square root of 8 by 3 - 1 by 3, x1 is anyway getting squared. So, that will also be 2 by 3.

So in fact, we have 4 values and the theory of Lagrange multipliers can be used to say that you are function constrained by g(x)is equal to 0, we will attain the minimum at the point where one of these potential solutions is the smallest. In this case, it so happens that these 2 points have the smaller value, which is 2 by 3. So, if you compare 6 to 2 by 3 and 2 by 3, 2 by 3 seems is the smaller one. So, and that is achieved 2 points and both these points happen to be points where the function attains its minimum value in given the constraint set.

And the point where you have the highest value in this case 6 happens to be the point where you can show that this is the point where the function will attain its maximum value also. So which means that these 2 are minimizers and in this case, this guy happens to be in the maximizer and this happens to be what is called a saddle point.

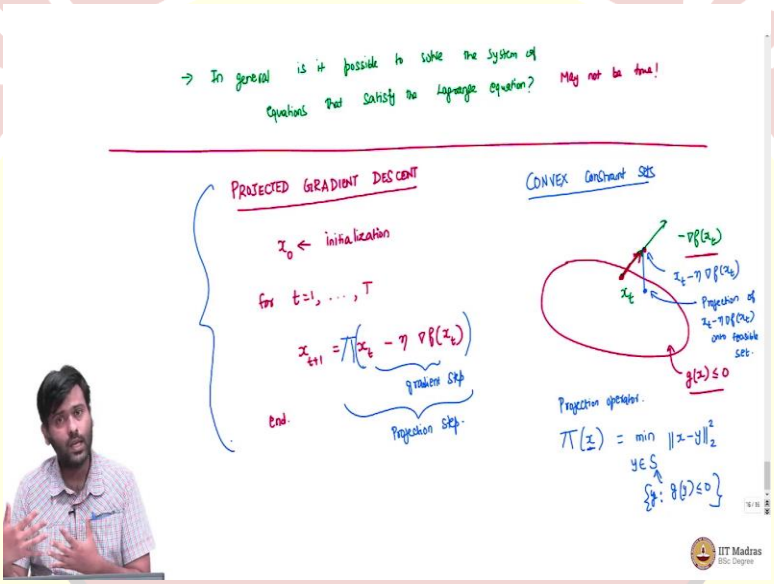
So, this is neither a minimizer nor a maximizer for this function. So, but let us leave at a side. So, the idea was, so let me summarize what we did in terms of the method of Lagrange multipliers. What we did was the following. We started by saying that if x star is an optimal solution, then it necessarily must be the case that the Lagrange equation is satisfied and the constraint is satisfied and we converted that by looking at the gradients into a set of equations.

And by carefully analysing these equations, what we did was, we kind of solved for λ and use that to solve for x1 and x2. So, and that gave us in this case 4 possibilities and by is plugging in each of these possibilities into the function, we saw that some of these points actually give the smaller

values and some of these points give larger values and the points which give the minimum values are indeed the minimizers of this function and the point that gives the higher values are actually end up being the maximizers of this function.

So, this in general is the method of Lagrange multipliers, where you can directly solve this problem without really moving in any direction or doing any iterative process or anything of that sort. So, this is a very good method when you can actually solve this solve for λ and on the equations.

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So, the next question that brings us to our next question that, like in general, is it possible to solve the system of equations that Lagrange that satisfy the Lagrange equation? In general, this may not be true. So, may not be true. It is not always possible to solve your problem analytically, like how we did for the simple example.

This is nice example to illustrate the power of Lagrange multipliers. But in general, it may not be true. So now, what, what else can we do? When we do not have an easy way to solve using the method of Lagrange multipliers, can we do something else. Like, kind of like gradient descent which will iteratively get us to the optimal solution. There are several possible ways of attacking this problem.

What I want to do today is give you a method for now and then we will talk more about when this method would actually do perform well. Under what conditions can we argue that the method of the algorithm that I am going to give now we will actually get to the optimal solution. So, the

method of Lagrange multipliers is very powerful, but then it needs careful solving of equations and so on.

But then we still want a method that will automatically get us to the optimal and, and gradient descent was one such method, but then it worked only for constraint, unconstrained optimization problems, what is the can we do something for constrained optimization problem, maybe we will have to sacrifice something with respect to the type of functions for which the method will work nevertheless, is there a method that will work for a general constrained optimization problem.

The answer is yes or no and I am going to put down the method and then we will talk about when this method will actually work a little later. The method is remarkably similar to the gradient descent algorithm. In fact, it is what is called as a projected gradient descent algorithm. So, what is the projected gradient descent algorithm? What does this algorithm do? This, what was let us remind ourselves what was gradient descent doing.

So, you start with some x naught so initialization and then for t equals 1 dot dot dot maybe there is upper bound, depending on how many number of iterations you want run this algorithm for, then you say that xt + 1 equals xt - 8 a times ∇f of xt the gradient at xt, this was our project, this was our simple gradient descent algorithm.

Now, what was the problem with this algorithm, the problem with this algorithm is this xt + 1, this update, when I am in sum xt, if I move, let us say this is my negative ∇, f of xt direction. If I move along this direction, it might so happen that if you take a step size and if I move here, my constraint set could be something like this.

So, this is the set of points where g(x)is let us say, less than or equal to 0 for instance. Now I am at sum xt, I my function actually grows in this particular direction, if I move in this direction, I might potentially take a step that will take me out of the feasibility set. Now, there is, it is very hard to tune step sizes to always remain within the visibility set because that is a property that will intimately depend on the property of the feasibility set and so on.

On the other hand, what we could potentially do is we will assume that, so I am at some point, I will take a gradient step, I might potentially go out of the gradient out of the feasible region, what I am going to do now is after taking this gradient step, I will basically come back to my feasible set by projecting the point where I end up after taking the gradient step on to the feasible set. So,

this point, let us call this this is my xt - eta ∇f of xt and now, this point is going to be the projection of xt - ∇f of xt on to feasible set.

So, we will call this projection. So, what is a projection? A projection is a function basically so which takes let us call that function pi. So, it takes any vector x and then it finds the minimum y such that the length squared is as small as possible, but what is the what is what should y satisfy it ca not be any y, the y should belong to some set S.

Now, what is the set S? We will say the set S is simply our constraint set which is the set g(x)is less than or equal to 0, insert of all or rather let me say y set of all y such that g of y is less than or equal to 0. So basically, from the red point, which you end up outside the feasible set, you find the closest point in the feasible set and move to that point and the operator that does that is what we will call as a projection operator. So, this is a projection operator.

So, all we are doing is that we are taking a gradient step, that might potentially push us outside the constraint region. So, we want to come back into the constraint region and what would be the best way to come back find that point which is closest in the constraint region and come there. So and so the only modification of gradient descent to projected gradient descent is that we have this projection operator.

So, this is the gradient step and this is the projection step. Now, this is an iterative algorithm. So, this is very much like our gradient descent algorithm, just that we are doing this extra projection step. The question is, this seems very simple. But then what is the catch? So, the catch is that there are a couple of things that one might be one should be very aware of here. One is, this depends on the fact that you can, in fact, do your projection onto your constraint set efficiently.

So, that is the first important thing. So, if your constraint set is really messy so it is really complicated, then doing this projection might not be that easy. So, but if you do have a constraint set, which is nice, where you can efficiently implement this projection operator, when I say efficiently, it means that let us say it is polynomial in the dimension of x, let us say, even if it is even, let us say if it is not polynomial, let us say maybe if it is faster than that, it is even better but then you want at least polynomial time efficiency for implementing this projection operation. If that that is possible for your constraint set, then this algorithm can be run.

Now, the question is, is there anything else that we should be aware of for guaranteeing that as I keep doing this, I would indeed converge to the optimal solution. So, I want to finally converge to a point where in the constraint set which minimizes my function. Of course, I will be within my constraint set, because by definition, I am doing this projection onto the constraints set which means that every iterate that I have, I will be inside my constraints.

But the question is, eventually, as I keep doing this will I eventually reached the, indeed the optimal point or not that is what we should be worried about and one can show and we will talk more about this in the next class in the next video, we will actually show that if your constraint set satisfy something nice in specifically, if your constraint set is a convex set.

So, and we briefly talked about convexity earlier also, it is a convex constraint sets and so it which means that it is in some sense, think of it as a nice set, we will define this formally next time, because it becomes important to understand convexity from a machine learning point of view.

We will talk more about convex sets in the next as we go along. But then, let me briefly tell you that if your constraint set is nice in the sense that it is convex, then we can argue that this algorithm will in fact, give you the optimal answer and if your function that you are optimizing is also convex and has some nice property, we can also argue how fast you will converge to the optimal, probably we will not do that in this set of lectures.

But what I also want you to understand is that the goodness of the algorithm also comes with respect to the function that you are trying to optimize and the set in which your constraint is in. So, as long as the set is convex, this algorithm will do well and if your function has more nice properties, some of which we will talk next time, this algorithm will do even better.

So, even better, meaning that of course, you will reach only the optimal point. So, there is nothing more with respect to the point that you will reach that you can do better, but then you will do better in terms of how fast you will reach that point. So that can be argued in this case.

So, this is the projected gradient descent algorithm and we will talk a bit about this in the next coming lectures and will for now, I want to summarize by saying that so far, we have looked at the problem of optimization in general we have talked about a gradient descent which works for unconstrained optimization problems, we looked at the constraint case and we came up with this method of Lagrange multipliers, we saw an example where you can in fact, analytically solve

certain problems using the method of Lagrange multipliers in some cases, you may not be able to solve it.

And so we want like a general purpose algorithm a iterate algorithm and when and it turns out that when you have nice functions and nice constraint sets, the projected gradient descent algorithm works wonders and we will see in a lot of machine learning applications, your functions are typically nice, your sets are typically nice. When I say nice, it means that they are convex and so these projected gradient type algorithms typically tend to work well. We will talk more about convexity and how it is related to machine learning in the coming classes. Thank you.

